Related Work

No.	Study	Dataset & Application	Method / GNN Type	Main Result	Strength	Limitation / Research Gap	Relation to Our Work
1	Zhou et al. (2025) — *Hand Gesture Recognition From sEMG Signals With GCN and Attention Mechanisms*	NinaPro DB2 and UOW; prosthetic-hand gesture recognition	CovGCN with covariance-based adaptive graph and attention	Outperformed several baseline models	Uses exact NinaPro DB2 and learns gesture-specific electrode relationships	No controlled comparison of anatomical, correlation, fully connected and adaptive graphs under one strict subject-independent protocol	Main competitor; motivates comparison of different graph structures on DB2
2	Lee et al. (2025) — *Decoding Gestures in Electromyography*	Five public EMG datasets, including NinaPro DB5	Spatial and spatiotemporal GCN using KNN/fully connected graphs and temporal edges	Strong intra- and inter-subject performance	Multi-dataset evaluation, graph comparison and interpretability	Does not evaluate the full 49-class, 12-channel NinaPro DB2 setting	Guides spatial/temporal graph design and electrode-importance analysis
3	Xu et al. (2024) — *Cross-User Electromyography Pattern Recognition Based on a Spatial-Temporal GCN*	HD-sEMG; 11 subjects and 17 gestures	CNN-MSTGCN and transfer-learning-based TL-CMSTGCN	68.0% user-independent and 92.3% after transfer learning	Focuses on cross-user recognition and statistical comparison	Uses HD-sEMG, fewer gestures and target-user data for the best result	Supports subject-wise splitting and calibration-free unseen-subject evaluation
4	Vipulanandan et al. (2024) — *A GNN Model for Real-Time Gesture Recognition Based on sEMG*	Private MyoBand dataset; 8 subjects and 8 electrodes	Muscle-activation graph with real-time GNN	99% accuracy and about 48 ms processing time	Demonstrates real-time feasibility	Small private dataset and unclear unseen-subject protocol	Motivates reporting inference time and computational efficiency
5	Xu and Jiang (2023) — *A Hybrid Model Based on ResNet and GCN for sEMG Gesture Recognition*	NinaPro DB1; hand-gesture recognition	ResNet-GCN joint-voting model	90.07% accuracy	Combines local signal features with muscle-synergy graph features	Uses DB1 and does not isolate the contribution of the graph component	Motivates pure GNN versus CNN-GCN comparison and ablation
6	Wang et al. (2025) — *Dynamic Graph Transformer-Convolutional Network for Multi-Channel EMG Gesture Recognition*	Public multichannel EMG dataset	Dynamic GCN with Transformer	97.24% accuracy	Models time-varying channel relationships and long-term dependencies	Complex architecture and limited evaluation details; not validated on NinaPro DB2	Motivates static-versus-dynamic graph comparison after simpler GNN baselines

References

[1] H. Zhou et al., "Hand Gesture Recognition From Surface Electromyography Signals With Graph Convolutional Network and Attention Mechanisms," *IEEE Sensors Journal*, vol. 25, no. 5, pp. 9081–9092, 2025. doi: 10.1109/JSEN.2025.3527918.

[2] H. Lee et al., "Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 404–419, 2025. doi: 10.1109/TNSRE.2024.3523943.

[3] M. Xu et al., "Cross-User Electromyography Pattern Recognition Based on a Novel Spatial-Temporal Graph Convolutional Network," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 72–82, 2024. doi: 10.1109/TNSRE.2023.3342050.

[4] P. Vipulanandan et al., "A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals," in *Proceedings of IEEE EMBC*, 2024. doi: 10.1109/EMBC53108.2024.10781990.

[5] X. Xu and H. Jiang, "A Hybrid Model Based on ResNet and GCN for sEMG-Based Gesture Recognition," *Journal of Beijing Institute of Technology*, vol. 32, no. 2, pp. 219–229, 2023. doi: 10.15918/j.jbit1004-0579.2022.116.

[6] P. Wang et al., "Dynamic Graph Tranformer-Convolutional Network for Multi-Channel EMG Gesture Recognition," in *Proceedings of IEEE EMBC*, 2025. doi: 10.1109/EMBC58623.2025.11253459.